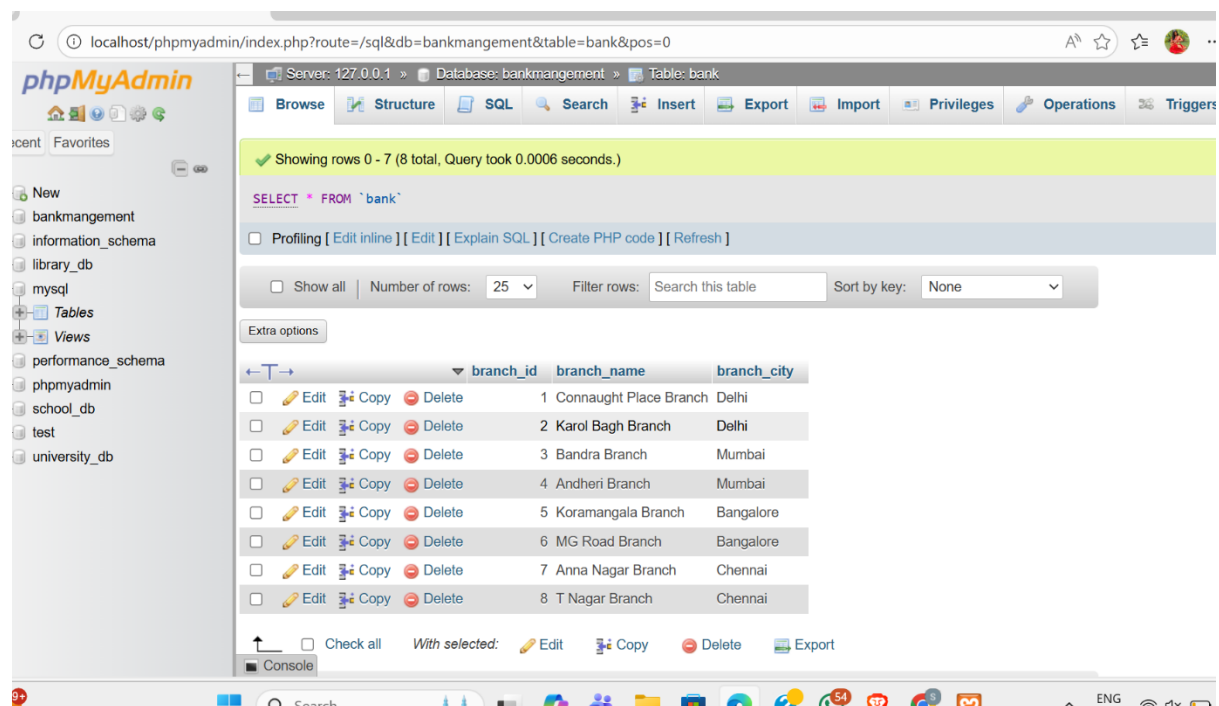


Dbms assessment

Bank table

```
CREATE TABLE bank (  
    branch_id SERIAL PRIMARY KEY,  
    branch_name VARCHAR(100) NOT NULL,  
    branch_city VARCHAR(100) NOT NULL  
);
```



Account Holder Table

```
CREATE TABLE account_holder (  
    account_holder_id SERIAL PRIMARY KEY,  
    account_no VARCHAR(20) UNIQUE NOT NULL,  
    account_holder_name VARCHAR(100) NOT NULL,  
    city VARCHAR(100) NOT NULL,  
    contact VARCHAR(15),
```

```

date_of_account_created DATE DEFAULT CURRENT_DATE,
account_status VARCHAR(20) CHECK (account_status IN ('active', 'terminated')),
account_type VARCHAR(50),
balance DECIMAL(15, 2) DEFAULT 0.00
);

```

The screenshot shows the phpMyAdmin interface for a database named 'bankmanagement'. The 'account_holder' table is selected, and the 'Browse' tab is active. The table contains 8 rows of data. The status of the accounts is as follows:

account_holder_id	account_no	account_holder_name	city	contact	date_of_account_created	account_status
1	ACC001	Rajesh Kumar	Delhi	9876543210	2024-01-10	active
2	ACC002	Priya Sharma	Delhi	9876543211	2024-01-18	active
3	ACC003	Amit Patel	Mumbai	9876543212	2024-02-05	active
4	ACC004	Sneha Reddy	Mumbai	9876543213	2024-02-20	active
5	ACC005	Vikram Singh	Bangalore	9876543214	2024-03-12	active
6	ACC006	Anjali Nair	Bangalore	9876543215	2024-03-25	terminated
7	ACC007	Suresh Iyer	Chennai	9876543216	2024-04-16	active
8	ACC008	Kavitha Menon	Chennai	9876543217	2024-04-08	active

Loan Table

```

CREATE TABLE loan (
    loan_no SERIAL PRIMARY KEY,
    branch_id INT REFERENCES bank(branch_id),
    account_holder_id INT REFERENCES account_holder(account_holder_id),
    loan_amount DECIMAL(15, 2) NOT NULL,
    loan_type VARCHAR(50)
);

```

phpMyAdmin

Recent Favorites

Server: 127.0.0.1 » Database: bankmangement » Table: loan

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Showing rows 0 - 7 (8 total, Query took 0.0002 seconds.)

SELECT * FROM `loan`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	loan_no	branch_id	account_holder_id	loan_amount	loan_type
☐ Edit Copy Delete	1	1	1	400000.00	Home Loan
☐ Edit Copy Delete	2	2	2	250000.00	Education Loan
☐ Edit Copy Delete	3	3	3	50000.00	Personal Loan
☐ Edit Copy Delete	4	4	4	800000.00	Business Loan
☐ Edit Copy Delete	5	5	5	1200000.00	Home Loan
☐ Edit Copy Delete	6	6	6	300000.00	Car Loan
☐ Edit Copy Delete	7	7	7	450000.00	Education Loan
☐ Edit Copy Delete	8	8	8	600000.00	Personal Loan

Click to sort results by this column.
Shift + Click to add this column to ORDER BY clause or to toggle ASC/DESC.
Ctrl + Click or Alt + Click (Mac: Shift + Option + Click) to remove column from ORDER BY clause

Check all With selected: Edit Copy Delete Export

localhost/phpmyadmin/index.php?route=/sql&db=bankmange...

INTRA-BANK TRANSFER TRANSACTION

BEGIN;

UPDATE account_holder

SET balance = balance - 100

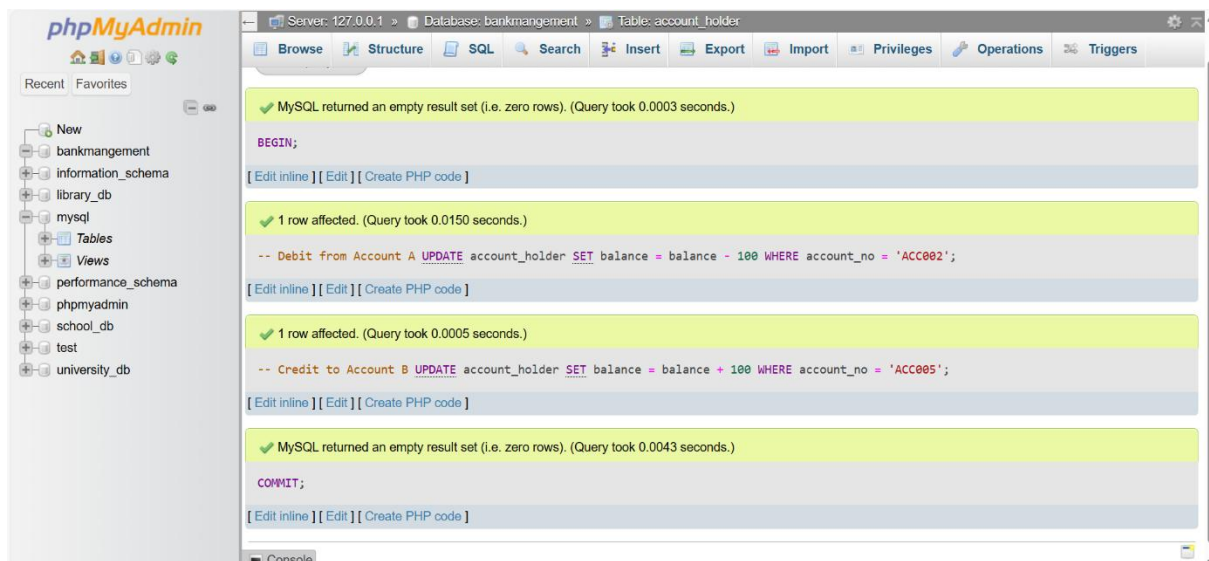
WHERE account_no = 'A';

UPDATE account_holder

SET balance = balance + 100

WHERE account_no = 'B';

COMMIT;



Queries

1. Fetch account holders from the same city

```
SELECT a1.account_holder_name, a1.city, a1.account_no
FROM account_holder a1
WHERE a1.city IN (
    SELECT city
    FROM account_holder
    GROUP BY city
    HAVING COUNT(*) > 1
)
ORDER BY a1.city, a1.account_holder_name;
```

The screenshot shows the phpMyAdmin interface with the 'account_holder' table selected. The SQL query is entered in the SQL tab, and the results are displayed in a table. The query filters for cities with more than one account holder and orders the results by city and account holder name.

Showing rows 0 - 7 (8 total, Query took 0.0008 seconds.) [city: BANGALORE... - MUMBAI...] [account_holder_name: ANJALI NAIR... - SNEHA REDDY...]

```
SELECT a1.account_holder_name, a1.city, a1.account_no FROM account_holder a1 WHERE a1.city IN ( SELECT city FROM account_holder GROUP BY city HAVING COUNT(*) > 1 ) ORDER BY a1.city, a1.account_holder_name;
```

Extra options

	account_holder_name	city	account_no
☐	Anjali Nair	Bangalore	ACC006
☐	Vikram Singh	Bangalore	ACC005
☐	Kavitha Menon	Chennai	ACC008
☐	Suresh Iyer	Chennai	ACC007
☐	Priya Sharma	Delhi	ACC002
☐	Rajesh Kumar	Delhi	ACC001
☐	Amil Patel	Mumbai	ACC003
☐	Sneha Reddy	Mumbai	ACC004

2. Accounts created after 15th of any month

```
SELECT account_no, account_holder_name  
FROM account_holder  
WHERE EXTRACT(DAY FROM date_of_account_created) > 15;
```

The screenshot shows the phpMyAdmin interface for a database named 'bankmanagement'. The 'Table: account_holder' is selected. The SQL query executed is: `SELECT account_no, account_holder_name FROM account_holder WHERE EXTRACT(DAY FROM date_of_account_created) > 15;`. The results show 4 rows (0-3) with the following data:

account_no	account_holder_name
ACC002	Priya Sharma
ACC004	Sneha Reddy
ACC006	Anjali Nair
ACC007	Suresh Iyer

The interface also shows a 'Console' tab at the bottom with the text 'sults operations'.

3. City name with branch count (alias: Count_Branch)

```
SELECT branch_city, COUNT(*) AS Count_Branch  
FROM bank  
GROUP BY branch_city;
```

phpMyAdmin

Recent

Favorites

New

bankmanagement

information_schema

library_db

mysql

Tables

Views

performance_schema

phpmyadmin

school_db

test

university_db

Server: 127.0.0.1 » Database: bankmanagement » Table: bank

Browse

Structure

SQL

Search

Insert

Export

Import

Privileges

Operations

Triggers

Showing rows 0 - 3 (4 total, Query took 0.0006 seconds.)

SELECT branch_city, COUNT(*) AS Count_Branch FROM bank GROUP BY branch_city;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all

Number of rows: 25

Filter rows: Search this table

Extra options

branch_city	Count_Branch
Bangalore	2
Chennai	2
Delhi	2
Mumbai	2

Show all

Number of rows: 25

Filter rows: Search this table

Query results operations

Print

Copy to clipboard

Export

Display chart

Create view

4. Account holders with loans (using JOIN)

SELECT

ah.account_holder_id,

ah.account_holder_name,

l.branch_id,

l.loan_amount

FROM account_holder ah

JOIN loan l

ON ah.account_holder_id = l.account_holder_id;

The screenshot shows the phpMyAdmin interface for a database named 'bankmanagement'. The 'Table: bank' is selected. The SQL query entered is: `SELECT ah.account_holder_id, ah.account_holder_name, l.branch_id, l.loan_amount FROM account_holder ah JOIN loan l ON ah.account_holder_id = l.account_holder_id;`. The query results are displayed in a table with 8 rows. The table has columns: account_holder_id, account_holder_name, branch_id, and loan_amount. The results are as follows:

account_holder_id	account_holder_name	branch_id	loan_amount
1	Rajesh Kumar	1	500000.00
2	Priya Sharma	2	250000.00
3	Amit Patel	3	150000.00
4	Sneha Reddy	4	800000.00
5	Vikram Singh	5	1200000.00
6	Anjali Nair	6	300000.00
7	Suresh Iyer	7	450000.00
8	Kavitha Menon	8	600000.00

The interface also shows a sidebar with a tree view of databases and tables, including 'bankmanagement', 'account_holder', 'bank', and 'loan'. The bottom of the interface has a 'Query results operations' section with buttons for 'Console', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'.